

SYNOPSIS OF THE B.TECH PROJECT ENTITLED

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**A synopsis Submitted
In Partial Fulfillment of the Requirements
for the Degree of**

**BACHELOR OF TECHNOLOGY
(Specialization name)**

By

Student name

Enroll. No.-

Under the Supervision of

Supervisor name

Designation



AMITY UNIVERSITY PUNJAB, INDIA

Declaration

I hereby declare that this SYNOPSIS report entitled, “ **Title Name**”, being submitted by me to **Amity University Punjab** for the award of the degree of Bachelor of Technology (**Specialization name**) is a record of bonafide work carried by me (Roll No.-.....). I worked under the guidance and supervision of **Supervisor name & Designation**.

The matter embodied in this report has **not** been submitted to any other university or institute earlier for the award of any degree or diploma.

Student Name

B. TECH. Sem (Session year)

ROLL NO.-

Acknowledgements

I have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. We would like to extend my sincere thanks to all of them.

I would like to take the opportunity to express my humble gratitude to **Supervisor name & Designation.** under whom I executed this project and most thankful his constant guidance and willingness to share his vast knowledge made me understand this project and its manifestation in great depth and helped me to complete the assigned task.

PLACE:

DATE:

Student Name

B. TECH. Sem (Session year)
ROLL NO.-

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Introduction

TechDec is a Online Technical Discussion Forum who provides the Technical solution of your problems.

This “**TechDec**” is aimed at developing online form for the group discussion. This is a web-based application to manage group discussion forum. Whenever a question is asked by the end-user to get information ,it is received by the administrator. Any user can post the doubts topics and can reply for the other user doubts. There is a centralized database in which all the information is managed. The user can invites others for Discussion and submit query. The administrator has the rights to update the database. This is useful for a small office, school or a department or for that matter any group who is interested to organize it effectively. There are also connected user who acts as an intermediate user who can also answer the questions of the end-user if they know it. Facility to share the resource and post articles that can be viewed by registered user. Whenever the new information is arrived it is updated by the end-user.

This TechDec Website involves with three types of users / Modules are: -

- **Administrator**
- **Guest User**
- **TechDec User**

Objectives

- To automate the multiple answers of the problem that helps people to find out relevant solution.
- To automate the online discussion forum that helps users to easily find the answers & get your relevant solution.
- To help users to provide skilled candidates with the help of Scale Point System.
- To help the users to update their Profiles & do the Query.
- It reduces cost to buy many books to find the relevant Solution for any query.
- It reduces time to find the solutions & get many answers of their queries.
- TechDec always get updated queries for your solution of the problems via notifications.

Roles and Responsibilities

- Analysis users and Guest users module for TechDec
- Designed mock up UI for users and guest users module
- Implement view, model & controller for users and guest users module
- Designed tables w.r.t users and guest users module
- Fix issues while review by Project manager

Requirements Analysis

The requirement analysis is done in order to understand the problem the software system is to be solved. The emphasis in requirement analysis is on identifying what is need for the system not how will achieve its goal. The goal of requirement activities is to document requirement in a software requirement specification.

There are two major phase in this phase

1. Problem understood or analysis.
2. Requirement specification.

In problem analysis the aim is to understand the problem and its context, the requirement of the system to be develop. Once problem analysis done and requirements understood. Requirements must be specified in requirement specification document. The requirement document must be specifying all function and performance of requirement. Format of input and output and design constraints due to political economical and environmental and security reasons. In other word besides functional specification all the factor the affect and design and proper functionality of the system must be specify in requirement document. Overall in requirement analysis we identify the requirement and constraints of proposal system. We can define requirement as “A condition of capability by a user to solve a problem or achieve an objective”

“A condition or a capability that must be met or possessed by a system..... to satisfy a contract, standard, specification, or other formally imposed document.”

Software Specifications

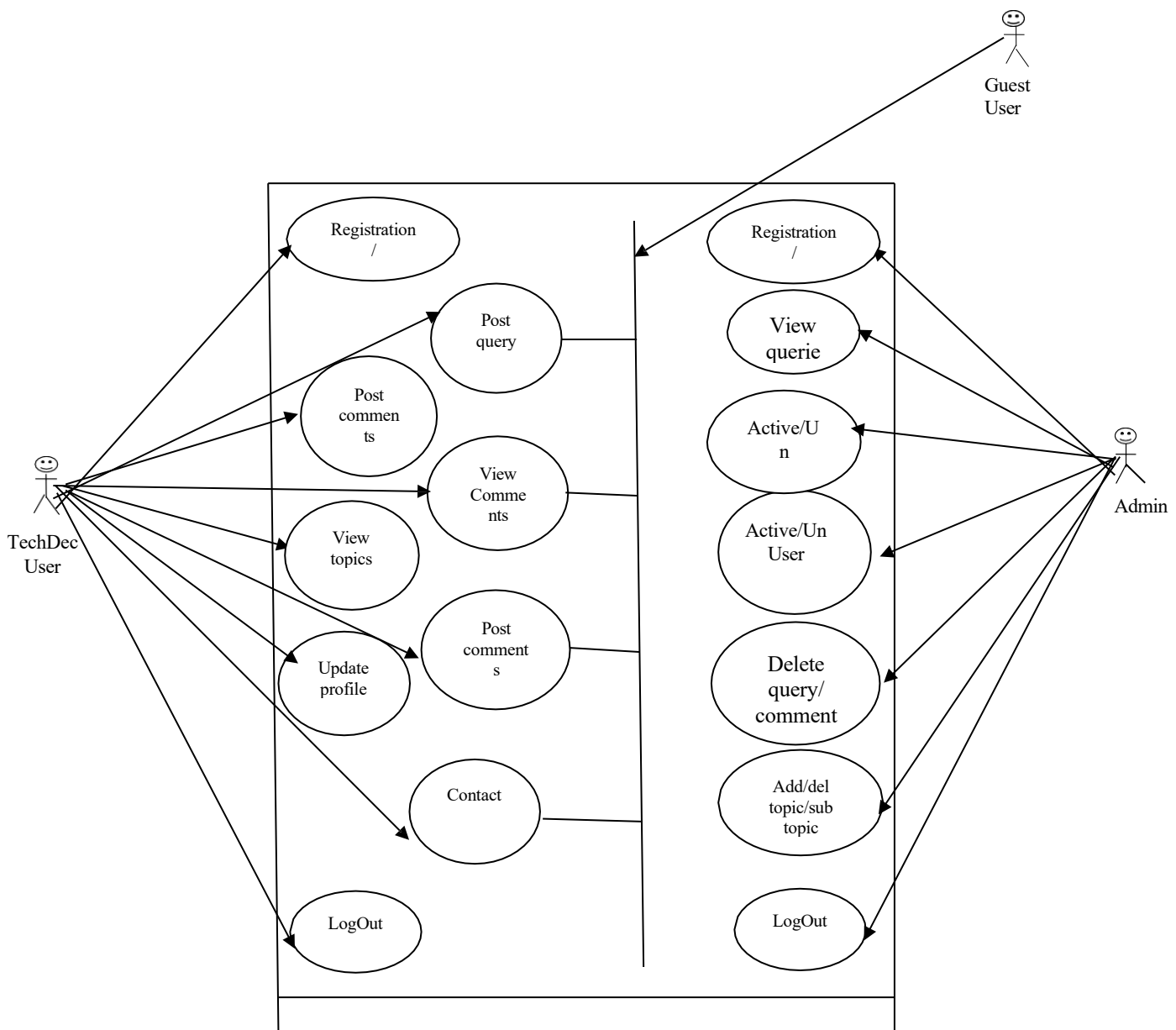
OS	Windows 7 & Linux
IDE& Other tools	Eclipse, Navicat, Beyond, Compare, Pencil
Versioning System	Tortoise SVN
Database	MySQL
Web Server	Tomcat 7.0
Server side Languages& other tools	Java 7, JDBC, JSP 2.2, Servlet 3.0.1
Client Side Languages & Other tools	HTML, XML, CSS, JavaScript, JQuery, Bootstrap, JSON, AJAX

System Design

System design is the process or art of defining the architecture, component, module, interface the data for a system to specified requirement. Object-oriented analysis and design (OOAD) methods are becoming the most widely used method for computer system design. The UML has become the standard language used in Object-oriented analysis and design. It is widely used for modeling Software System and is increasing used for high designing non-software system and organization.

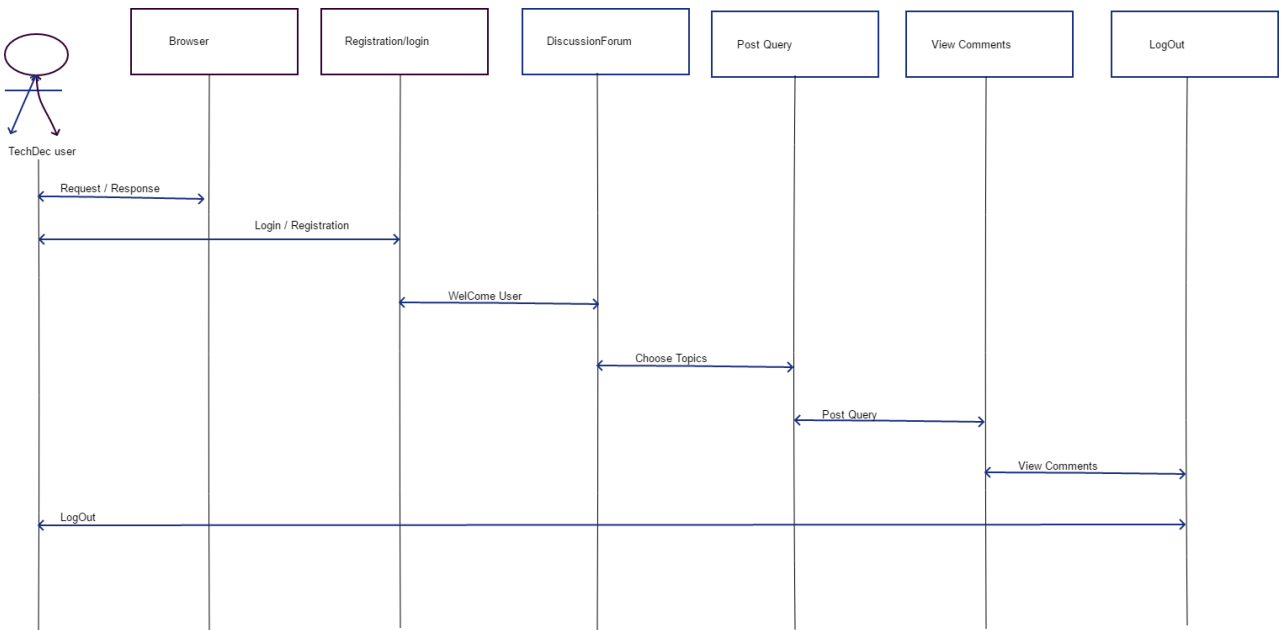
Use Case Diagram

A use case diagram at its simplest is a representation of a user's interaction with the system that shows the relationship between the user and the different use case in which the user is involved. A use case diagram can identify the different types of users of a system and the different use cases and will often be accompanied by other types of diagrams as well.



Sequence Diagram

A Sequence diagram is an interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Charts. A sequence diagram shows object interactions arranged in time sequence.



Data Flow Diagram

DFD (data flow diagrams) known as a bubble chart has the purpose of clarifying system requirement and identifying major transformation that will become programs in system design. So it is starting point of the design phase that functional decompose the requirement specification down to lowest level of details. A DFD consists of bubbles joined by the lines. The bubbles represent the data transformation and the lines represent data flow in the system.

DFD describes what data flow rather than how they are processed. The key question that we are trying to answer is: what major transformation must occur for input to be correctly transformed into output.

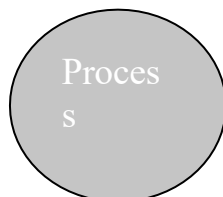
Data Flow Diagram Symbols

External Entity



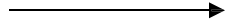
An external entity is a source or destination of a data flow. Only those entities which originate or receive data are represented on a business process diagram. The symbol used is a square containing a meaningful and unique identifier.

Process



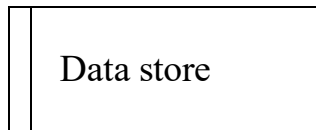
A process shows a transformation or manipulation of data flows within the system. A circle represents a process that transfer incoming data flow into outgoing data flow.

Dataflow



A data flow shows the flow of information from its source to its Destination. A data flow is represented by a line, with arrowheads Showing the direction of flow. Information always flows to or from a process and may be written, verbal or electronic. Each data flow may be referenced by the processes or data stores at its head and tail, or by a description of its contents.

Data store



A data store is a holding place for information within the system:
It is represented by an open ended narrow rectangle.

Steps to Construct Data Flow Diagrams

Four steps are commonly used to construct a DFD

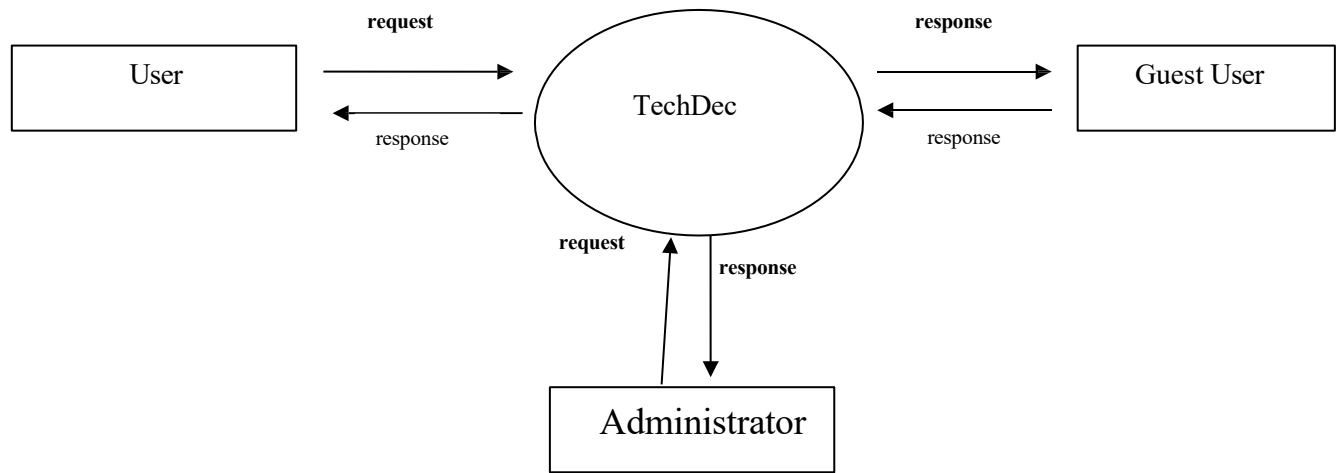
- Process should be named and numbered for easy reference. Each name should be representative of the process.

- The direction of flow is from top to bottom and from left to right.
- When a process is exploded into lower level details they are numbered.
- The names of data stores, sources and destinations are written in capital letters.

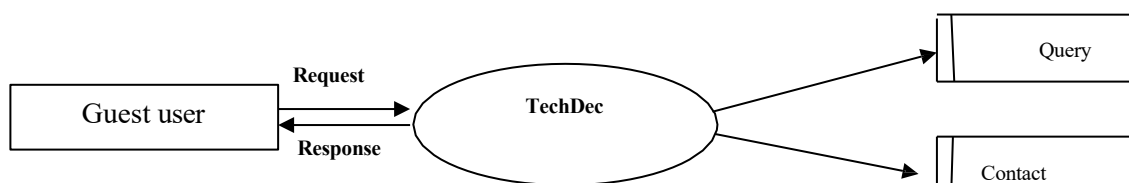
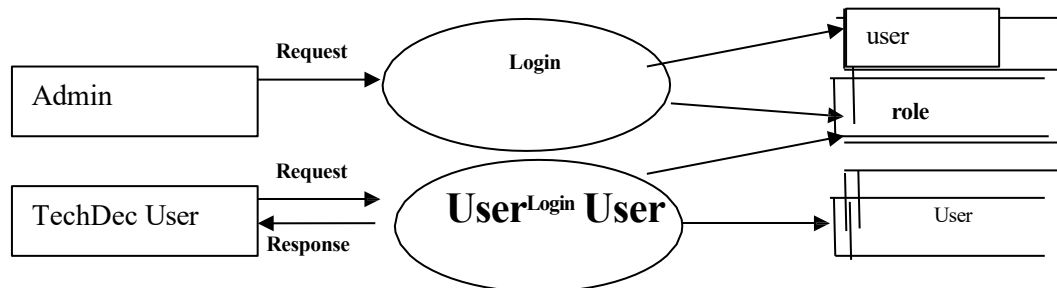
Rules for constructing a Data Flow Diagram

- Arrows should not cross each other.
- Squares , Circles and files must bear names.
- Decomposed data flow squares and circles can have same names. Choose meaningful names for dataflow.
- Draw all data flows around the outside of the Diagram.

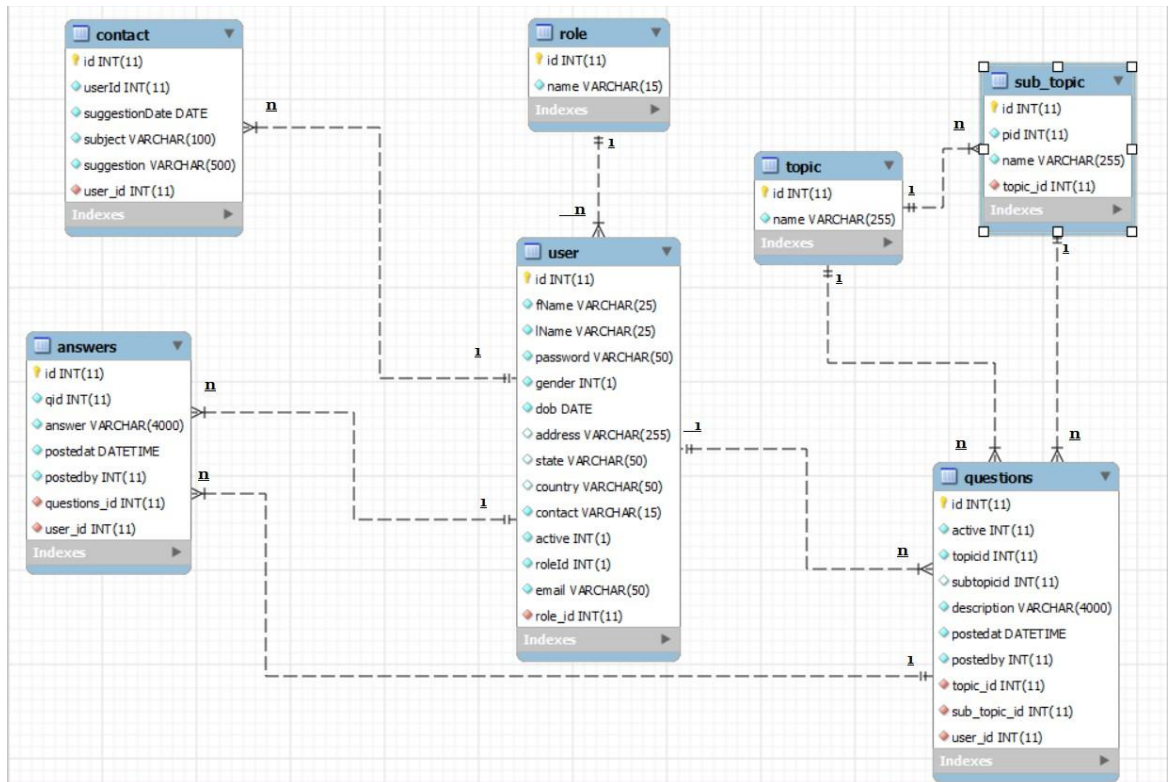
Level 0 DFD



Level 1 DFD



ER Diagram



TIME FRAME (GANTT CHART)

Gantt Chart Word Template

[illegible]

REFERENCES

- [1] T. Pitman, "The evolution of the student as a customer in Australian higher education: a policy perspective," *Aust. Educ. Res.*, vol. 43, no. 3, pp. 345-359, May 2016, doi: 10.1007/s13384-016-0204-9.
- [2] L. Baiazitova, J. Skopalik, V. Cmiel, J. Chmelik, O. Svoboda, and I. Provaznik, "Modern semi-automatic set-up for testing cell migration with impact for therapy of myocardial infarction," in *IFMBE Proc.*, May 2019, vol. 68, 3 ed., pp. 155-159, doi: 10.1007/978-981-10-9023-3_28.
- [3] J. Chapman, C. Landis, and S. Smith, "The development of a library Research Methods course for online graduate students in education," in *Information literacy programs in the digital age : educating college and university students online*, A. Daugherty and M. F. Russo Eds. Chicago, IL: Association of College and Research Libraries, 2007, ch. 3, pp. 13-22.
- [4] *Petroleum, petrochemical and natural gas industries - Composite repairs for pipework*, ISO 24817, ISO, Switzerland, 2017.
- [5] T. Liu *et al.*, "Biomimetic fabrication of new bioceramics-introduced fibrous scaffolds: From physicochemical characteristics to in vitro biological properties," *Mater. Sci. Eng., C*, vol. 94, pp. 547-557, Jan 2019, doi: 10.1016/j.msec.2018.09.063.
- [6] K. D. Patel, R. K. Singh, J. H. Lee, and H. W. Kim, "Electrophoretic coatings of hydroxyapatite with various nanocrystal shapes," *MatL*, vol. 234, pp. 148-154, Sep 2019, doi: 10.1016/j.matlet.2018.09.066.